

EmbedGuard Cross-Layer Detection Architecture

Detection Layers

Layer 1: Prompt Injection Detection

$$\beta_1 = 0.35$$

Layer 2: TEE Embedding Attestation

$$\beta_2 = 0.75$$

Layer 3: Retrieval Distributional Analysis

$$\beta_3 = 0.50$$

Layer 4: Output Consistency Verification

$$\beta_4 = 0.20$$

Threat Correlation Engine

ThreatScore = $\sum \beta_i \times \text{signal}_i$

Deployment

Modes
Passive Mode (Logging)

Gated Mode (Human Review)

Active Mode (Auto-Block)